

# BENCHMARKING SUSTAINABLE DEVELOPMENT PRACTICES IN KAZAKHSTAN'S HIGHER EDUCATION SYSTEM: EXPERIENCE OF REGIONAL UNIVERSITIES

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## ABSTRACT

Kazakhstan's modern higher education system is undergoing a transformation aimed at integrating sustainable development principles into educational, research, and management processes. The purpose of this study is to benchmark sustainable development practices at regional universities in Kazakhstan, using the Innovative University of Eurasia and Toraighyrov University in Pavlodar, Republic of Kazakhstan, as examples. The study is based on an analysis of strategic documents, program initiatives, curricula, and reporting data from the universities, as well as a comparative study of international models for implementing sustainable development in the university environment. The use of benchmarking methodology allowed us to identify similarities and differences in institutional approaches, identify strengths, and identify barriers and factors hindering the sustainable development of educational organizations. The results of the study demonstrate that both universities have significant potential in the areas of environmentally oriented management, digitalization of educational processes, and the development of sustainable development competencies in students. At the same time, the need to further strengthen the internal sustainability assessment system and expand partnerships with businesses and local communities is noted.

**Keywords:** sustainable development; higher education; benchmarking; university ecosystem; institutional development; educational strategies.

## INTRODUCTION

Current trends in the development of higher education worldwide demonstrate the growing role of universities in achieving the Sustainable Development Goals defined by the United Nations. Amid global challenges related to climate change, the digitalization of the economy, and the need to foster environmental responsibility, universities are becoming key agents of social and technological transformation. This increases the need to develop institutional governance models that integrate sustainable development principles into educational and research processes.

Kazakhstan, as a signatory to the UN 2030 Agenda, is taking active steps to adapt its national education

system to sustainable development objectives. Strategic documents of the Ministry of Science and Higher Education of the Republic of Kazakhstan outline the need to integrate the principles of environmental friendliness, social responsibility, and economic efficiency into university activities. However, the level of institutional maturity and consistency of these processes remains uneven, especially in regional universities where resources and infrastructure are limited. Regional universities, such as the Innovative University of Eurasia and Toraighyrov University (Pavlodar, Kazakhstan), are important centers for the development of human capital and innovative potential at the regional level. Studying their experiences in implementing sustainable development practices allows us to identify characteristics, trends, and challenges specific to the regional higher education sector in Kazakhstan.

The scientific novelty of this study lies in the application of benchmarking methodology to the analysis of institutional sustainable development practices at universities in Kazakhstan. This approach allows us not only to assess the current state of integrating sustainability principles into the universities' activities but also to identify areas for improvement based on comparisons with international examples.

The aim of the study is to conduct a comparative analysis of sustainable development practices at two regional universities in Kazakhstan and identify factors influencing the effectiveness of their implementation. The objectives include evaluating the universities' strategic documents and programs, analyzing educational initiatives aimed at developing sustainability competencies, and developing recommendations for strengthening the universities' institutional resilience. The results of the study have practical significance for the development of national and regional strategies for the sustainable development of higher education, as well as for the adaptation of international approaches to Kazakhstani conditions.

## LITERATURE REVIEW

Integrating sustainable development into higher education has become a key topic of international research. According to UNESCO, universities serve as centers for generating knowledge, competencies, and

values necessary for the implementation of the Sustainable Development Goals and act as agents of societal change [1]. Contemporary research emphasizes that the sustainable development of universities requires a shift from the traditional educational model to a comprehensive institutional transformation, encompassing governance, infrastructure, training, and interaction with external stakeholders [2].

The “whole-institution approach” model, proposed within the Higher Education for Sustainable Development program, views a sustainable university as a system in which principles of environmental, social, and economic responsibility permeate all levels of activity [3]. This approach is reflected in the international initiatives Global University Partnership for Environment and Sustainability and the Higher Education Sustainability Initiative, which have become platforms for exchanging experience and developing sustainability assessment criteria [4].

Universities in the European Union view sustainable development as a systemic component of strategic management. The European Commission, through the Erasmus+ and Horizon Europe programmes, supports the integration of sustainability principles into education and research [5]. A number of universities, such as the Technical University of Denmark and Utrecht University, have dedicated sustainability offices coordinating projects on carbon footprint reduction, energy efficiency, and inclusion in the academic environment [6].

In the United States, the Association for the Advancement of Sustainability in Higher Education (AASHE) plays a significant role, developing the Sustainability Tracking, Assessment, and Rating System (STARS), which enables benchmarking between universities and fosters a culture of responsibility for environmental impact [7].

In the context of Central Asia, Kazakhstan has demonstrated progress in adapting international approaches to the national education system. State documents—the “Concept of the Transition of the Republic of Kazakhstan to Sustainable Development by 2050” and the “State Program for the Development of Education and Science”—reinforce the country’s commitment to promoting the Sustainable Development Goals in education [8]. Research by Kazakhstani authors emphasizes that university sustainability is shaped by the intersection of three areas: institutional policy, educational initiatives, and partnerships with external stakeholders [9]. Regional universities, meanwhile, perform a special function: they connect academic research, local government, and business, shaping local models of sustainable development.

At the same time, the analysis reveals that most Kazakhstani universities are fragmented in their imple-

mentation of sustainable practices. Primary efforts are focused on the environmental component, while social and economic aspects remain less well-developed. Applying benchmarking in such a context allows us to assess the effectiveness of current practices and identify areas for improvement, providing a more comprehensive approach to sustainability management in higher education.

## METHODS

The study’s methodological framework is based on systemic, comparative, and institutional approaches, allowing for a comprehensive examination of sustainable development practices at regional universities in Kazakhstan. Benchmarking, aimed at identifying best practices and comparing university sustainable development indicators with national and international benchmarks, was chosen as the primary method. Benchmarking was used to shift from descriptive analysis to assessing the effectiveness of management, educational, and infrastructural strategies.

The study covered two leading regional universities in Pavlodar—Innovative University of Eurasia and Toraigrov University. The analysis was based on the universities’ strategic development plans, annual reports on the implementation of sustainable development programs, internal audit documents, educational programs, and official university websites. Additionally, government regulations governing the implementation of sustainability principles in higher education and international sources, including UNESCO, AASHE, and EUA methodological materials, were reviewed.

To assess the level of integration of sustainable development principles, a proprietary indicator system was used, comprising five blocks:

- strategic management and institutional policy;
- educational programs and competencies for sustainable development;
- research and innovation activities;
- campus infrastructure and environmental sustainability;
- interaction with society and business.

Each block was assessed on a scale of three levels: beginner, developing, and advanced. Content analysis and an expert survey of representatives of the academic community and university management personnel were used to verify the data.

The data obtained allowed us to identify the maturity of sustainable practices and determine areas for further development of university ecosystems, taking into account international standards and Kazakhstan’s realities.

**Results and Discussion.** The study results showed that the integration of sustainable development into

the activities of regional universities in Kazakhstan is in the active development stage. Despite a common regulatory framework and state strategic priorities, specific mechanisms for implementing sustainability principles vary depending on the institutional culture, human resource potential, and financial resources of universities.

An analysis of strategic documents and educational initiatives revealed five key areas for assessing sustainable development:

- Institutional policy and management strategy;
- Educational programs and the development of sustainable development competencies;
- Research and innovation activities;
- Environmental and infrastructural sustainability of the campus;
- Interaction with society and business.

1. Institutional policy and management strategy.

Both universities demonstrate a conscious shift from a declarative level to strategic management of sustainable development. InEU is implementing a project to introduce educational modules on sustainable development. Toraigyrov University has established a working group to implement the "Green University 2030" strategy, demonstrating the formation of an institutional framework for sustainability.

However, differences emerge in the level of integration: InEU emphasizes social and educational components, while Toraigyrov University devotes more attention to environmental and infrastructural aspects.

2. Educational programs and competencies for sustainable development. Both universities are seeing an increase in the number of disciplines related to sustainable development. However, these courses are currently optional and not always included in mandatory educational programs. InEU has made progress in creating master's programs that study environmental and economic management and social responsibility. Toraigyrov University is focused on integrating sustainability topics into engineering and agricultural specialties, reflecting regional needs.

For further development, it is necessary to expand student enrollment, integrate sustainability competencies into educational standards, and create a learning outcomes assessment system aligned with the Sustainable Development Goals.

3. Research and innovation activities. Research in the field of sustainable development is becoming an important area of university research policy. InEU is implementing projects on energy-efficient technologies, supported by international projects and a project from the Ministry of Science and Higher Education of the Republic of Kazakhstan. Toraigyrov University actively participates in international grants related to sustainable natural resource management. However,

the overall funding for such research remains limited, and the commercialization of its results is rare.

To strengthen scientific potential, it is necessary to stimulate the creation of interdisciplinary laboratories, form research consortia with enterprises, and develop a grant funding system.

4. Environmental and infrastructural sustainability of the campus. Both universities are demonstrating progress in developing environmental infrastructure. InEU is implementing projects to automate production processes, which has a positive impact on sustainable development and energy consumption. Toraigyrov University is actively developing renewable energy sources, installing solar panels and modernizing its heating and water supply systems.

These measures help reduce carbon emissions and foster environmental awareness among students. However, comprehensive solutions are needed in the areas of sustainable transportation management, water use, and digital monitoring of environmental indicators.

5. Interaction with society and business. Social sustainability is realized through active collaboration between universities and local authorities, businesses, and non-profit organizations. Both universities participate in sustainable urban development projects and organize environmental forums. InEU focuses on environmental education programs and support for school initiatives, while Toraigyrov University is developing corporate partnerships in the field of green technologies and energy efficiency.

Such forms of cooperation not only contribute to the practical implementation of sustainability principles but also foster a positive public image of universities as centers of regional development. The analysis showed that InEU and Toraigyrov University are at an advanced stage of developing their institutional sustainable development ecosystems. Key achievements include the development of strategic management, the environmental modernization of campuses, and partnerships with local communities. However, issues related to the systematic integration of sustainability principles into educational standards and the expansion of university participation in the commercialization of scientific research remain unresolved.

Benchmarking demonstrated that regional universities in Kazakhstan have significant potential for strengthening sustainability, provided they receive consistent government support, improve governance mechanisms, and develop international academic ties.

Conclusion. The results of the study confirm that the development of sustainable practices in Kazakhstan's universities is a prerequisite for improving the effectiveness of the national higher education system and its integration into the international academic space.

Benchmarking demonstrated that regional universities are capable of creating their own institutional models of sustainable development that reflect both global trends and the specifics of Kazakhstan's socio-economic environment.

Innovative Eurasian and Toraigyrov Universities demonstrated significant progress in strategic management, environmental modernization of campuses, and strengthening partnerships with the local community and business. These results confirm that sustainable development is possible not only at the level of leading national universities but also at regional universities, provided a systematic approach and institutional support are in place.

Nevertheless, the analysis revealed a number of challenges hindering the integration of sustainable development. These include limited funding, insufficient coordination between government and academic structures, fragmented educational initiatives, and a weak link between science and the economy. Addressing these issues requires the development of a national system for assessing university sustainability, based on unified indicators and performance criteria adapted to Kazakhstan's realities.

Practical recommendations include the institutionalization of sustainability centers at universities, the expansion of faculty training and professional development programs, the development of partnerships with industrial and environmental organizations, and the promotion of university participation in international sustainability networks and rankings.

Thus, the study's results can serve as a basis for improving public policy in education and science, as well as for formulating sustainable development strategies for Kazakhstan's universities, focused on the long-term social and environmental well-being of the country.

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